AGEL J020613−011417 — Drafting Fact Sheet for the Methods & Results Sections

Facts, decisions, and provenance keyed to the paper outline — last revised 2026-06-08 (post-M12)

Compiled for R. Córdova Rosado from project notes, memory, METHODS_AND_SYSTEMATICS.md, and TESTS_AND_DIAGNOSTICS.md

2026-06-08

How to use this document

This is a drafting reference, not prose for the paper. Every bullet is a fact or decision we have already made, with its source file in `[brackets]` so you can verify it. Where a number is superseded, the current value is given and the stale one flagged. The **<!-- PV:auto:… -->** blocks (the headline numbers below + the §3.1 budget table) are generated from **`results/PAPER_VALUES.json`** — do not hand-edit inside the markers; refresh with `python scripts/paper_values.py --render DRAFTING_FACTS_paper_2026-05-29.md`. Three gaps are flagged up front — they are not in this repo and you must source them elsewhere before drafting those paragraphs:

- G1 — Lens modeling (Ferrami et al.). No BPL / point-mass / M(<r_break) / joint-posterior / PSF / shear / subhalo content exists in this kinematics+photometry repo. Source from the lens-modeling repo (`~/Documents/AGEL/202509_DESJ0206_modeling/` or `20251112_DESJ0206_Pyauto_PRONTO/`) or the Ferrami draft directly.
- G2 — X-ray / "truly quiescent" statement. No X-ray, Chandra, or explicit quiescence data anywhere in the read files. The galaxy is only ever described as a "passive elliptical." You need an external source if you want the X-ray quiescence claim in §Results.
- G3 — "We do not account for peculiar velocities." This sentence is not currently written in any methods doc. It is a true statement about our analysis (we use the line-fit systemic z directly) — just add it; nothing to reconcile.

Headline numbers — *generated from `results/PAPER_VALUES.json` by `scripts/paper_values.py --render`; do not hand-edit inside the markers.*

- σ_e(<R_e) = 267.31 ± 12.79 km/s (asym −12.98 / +12.61); often rounded 267 ± 13 km/s. Measured at the best-mask R_e=2.097″ aperture; the R_e-source systematic (±5.13, best-mask CoG family) is folded into the budget (§3.1).
- log(M★/M⊙) = 11.47 +0.09/−0.14 (stat, 10%) — aperture-corrected total at the matched 2 R_e = 4.19″ elliptical aperture (empirical aperture + single-Sérsic model wings; GAMA/Taylor+2011 fluxscale). Five estimators: raw 11.18 (empirical) · raw+apcorr 11.35 · filled 11.36 · total 11.47 (headline) · Sérsic-total 11.40. Sérsic-only systematic ±0.19 dex; masking-approach ±0.09 dex. See §2.1.1b / §3.2.
- R_e = 2.097″ = 14.76 kpc (best-mask F140W+F200LP CoG; method systematic ±0.10″). Supersedes 2.305″ (expert mask).
- z_deflector = 0.67564; z_source = 1.302.

Cosmology: FlatLambdaCDM, H_0 = 70 km/s/Mpc, Ω_m = 0.3; 1″ ≈ 7.04 kpc (7043.5 pc/arcsec), D_A = 1452 Mpc at z = 0.67564. [reference_hst_jwst_data_properties.md]

> Staleness warning: older docs (`METHODS_AND_SYSTEMATICS.md` 2026-05-18 body; pre-2026-06-11 handoffs) print superseded σ_e numbers (254.85 / 268.98 / 269.62, totals ±18/±13.27). The current

headline is 267.31 ± 12.79 (post "best mask throughout", 2026-06-11, R_e=2.097″). Always quote the current value (canonical source: `results/PAPER_VALUES.json`).

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§1 Data and observations

All values below are read directly from the FITS headers of the analysis files (verified 2026-06-08), not from memory. Target: AGEL J020613−011417 = DES J0206−0114, ICRS RA = 31.55611°, Dec = −1.23817° (HST `RA_TARG/DEC_TARG`); deflector z = 0.67564, source z = 1.302. The field is the massive cluster ACT-CL J0206.2−0114 (ACT DR5, Hilton et al. 2021) — the JWST pointing target.

Facility	Inst. / mode	Band(s)	Program (PI)	UT date	Exp.	Pixel scale
Keck II	KCWI — KCRM red (RL) + KCB blue (BL), Medium slicer, 2×2	RL 5625–8941 Å (σ_e); BL λc 4500 Å	U204 + K409 + U002 (3-night combine)	2024-12 → 2025-11	≈180 min RED + ≈198 min BLUE (36 × 300 s RED)	0.300″/spaxel
HST	WFC3/UVIS	F200LP	16773 (Glazebrook)	2022-07-14	600.0 s (NDRIZIM 4)	0.050″/pix
HST	WFC3/IR	F140W	16773 (Glazebrook)	2022-07-14	597.7 s (NDRIZIM 3)	0.080″/pix
JWST	NIRCam SW, module B	F150W2 (CLEAR)	05594 (Mahler)	2024-08-27	1836.0 s	0.03075″/pix
JWST	NIRCam LW, module B	F322W2 (CLEAR)	05594 (Mahler)	2024-08-27	1836.0 s	0.06301″/pix

§1.1 Keck/KCWI IFU spectroscopy (the σ_e data)

- Keck II / KCWI, KCRM red arm, RL (Red-Low) grating (`RGRATNAM`), Medium slicer (`IFUNAM`), 2×2 binning (`BINNING/CCDSUM`); nod-and-shuffle off (`RNASNAM='Open'`).
- The headline cube is a 3-NIGHT, 3-PROGRAM combine (see §2.2.1), NOT a single dataset — the cube header (`DATE-OBS` 2025-11-17, `PROGPI` Jones, `XPOSURE` 300 s) describes only the first input frame, so cite the combine, not the header:

Night (UT)	Program (PI)	RED (RL)	BLUE (BL)
2024 Dec 29	U204	4 × 300 s	1 × 1320 s *(excluded by reducer)*
2025 Aug 30	K409 (PI TBD)	12 × 300 s	4 × 990 s
2025 Nov 17	U002 (Jones)	20 × 300 s	5 × 1320 s

  → 36 × 300 s ≈ 180 min RED + ≈ 198 min BLUE on-source. The NEW **`_mtwdo_`** reduction (Master-Twilight + Dome hybrid flats, K. R. Gupta), `Nov17_2025_DESJ0206_RL_combined_mtwdo_icubes_wcs.fits`; dithered (PA 0°/45°/90°); CR rejection via astroscrappy (`fsmode=median`, `psffwhm=2.50`). `OBSERVER` = Glazebrook, Gupta, Jones, Kacprzak, Alcorn, Rhoades, Barone, Tran, Chen, Vasa. *(K409 PI + Aug-30 DIMM seeing still TBD for acknowledgements.)*
- Both KCWI arms recorded simultaneously. The red arm (KCRM, RL grating, central λ 7150 Å `RCWAVE` — the cube below) is the σ_e dataset. The blue arm (KCWI/KCB, BL (Blue-Low) grating `BGRATNAM`, central λ 4500 Å `BCWAVE`, KBlue filter `BFILTNAM`, N&S off) is the ≈198 min BLUE above; it is not used for σ_e (the deflector's Ca H+K…Mg b absorption falls in the red) but supplies the source-z = 1.302 cross-check (Fe II UV multiplet; §2.2.2). The blue combined cube is not in this repo (only the RL cube) — quote the blue arm from the observing log, not a local file.

- Conditions (Nov-17 frame, header-literal): airmass 1.165 (`AIRMASS`); guide-star FWHM 1.271″ (`GUIDFWHM`) — the adopted seeing; parallactic angle 15.6°, rotator 0°. (Per-night airmass/seeing for the other two nights not carried in the combined cube.)
- Cube: 100 × 100 spaxels × 3317 λ-pixels; 0.300″/spaxel (square; from WCS), 30″ × 30″ FOV; wavelength 5625–8941 Å at 1.0 Å/pix (`CRVAL3/CD3_3`), red central λ 7150 Å (`RCWAVE`). Spectral resolution R ≈ 10000 at the fit band → σ_inst ≈ 12.6 km/s (well below σ_e ≈ 270; the LSF is carried into ppxf). [reference_kcwi_data_properties.md]
- Pointing `RA/DEC` = 02:06:13.38 / −01:14:20.8; target `TARGRA/TARGDEC` = 02:06:13.58 / −01:14:19.8.

§1.2 HST/WFC3 imaging (F200LP, F140W)

- Program 16773, PI Karl Glazebrook (`PROPOSID/PR_INV_*`); `TARGNAME` = DESJ0206-0114; both bands UT 2022-07-14 (`EXPSTART` MJD 59774.30–59774.31), ORIENTAT 111.9°, drizzled, units ELECTRONS/S.
- F200LP — WFC3/UVIS, aperture UVIS2: EXPTIME 600.0 s, NDRIZIM 4, NCOMBINE 2, drizzle scale 0.050″/pix; `PHOTFLAM` = 5.1851×10^{-20}, `PHOTPLAM` = 4923.48 Å → ZP_AB = 27.344.
- F140W — WFC3/IR: EXPTIME 597.7 s, NDRIZIM 3, NCOMBINE 3, drizzle scale 0.080″/pix; `PHOTFLAM` = 1.4829×10^{-20}, `PHOTPLAM` = 13922.91 Å → ZP_AB = 26.446.
- AB zeropoints are recomputed per-image from `PHOTFLAM/PHOTPLAM` (never hardcoded); `ZP_AB = −2.5·log10(PHOTFLAM) − 21.10 − 5·log10(PHOTPLAM) + 18.6921`. Drizzle pixel correlation: empirical noise ≈1.3–2× the CCD-equation value → motivates the 10% flux floor (§2.1.4).

§1.3 JWST/NIRCam imaging (F150W2, F322W2)

- Program 05594 — *"JWST Cluster SLICE — Strong Lensing and Cluster Evolution"*, PI Guillaume Mahler (SURVEY category); `TARGPROP` = ACT-CL_J0206.2−0114 (the cluster field; the deflector sits in it). Both bands UT 2024-08-27 (`DATE-BEG` 06:02–06:40), NIRCam module B, units MJy/sr (i2d), NINTS 1 / NGROUPS 4.
- F150W2 (SHORT channel, CLEAR pupil): `EFFEXPTM/XPOSURE` 1836.0 s; `PIXAR_SR` = 2.2222×10^{-14} sr → 0.03075″/pix (PIXAR_A2 = 9.454×10^{-4} ″^{2}) → ZP_AB = 28.033.
- F322W2 (LONG channel, CLEAR pupil): `EFFEXPTM/XPOSURE` 1836.0 s; `PIXAR_SR` = 9.3307×10^{-14} sr → 0.06301″/pix (PIXAR_A2 = 3.970×10^{-3} ″^{2}) → ZP_AB = 26.475.
- JWST AB from area: `ZP_AB = −6.10 − 2.5·log10(PIXAR_SR)`. No PAM correction (drizzled/resampled).

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§2 Methods

§2.1 Photometric Analysis

§2.1.1 Masking

- Use the F200LP **`_mask.fits`** as the arc mask — NOT the F140W one. [feedback_masking_strategy.md; reference_hst_jwst_data_properties.md; METHODS §II.4]
  - F200LP mask (2512 HST pixels) is hand-tuned to cover the arc only and leaves the deflector core unmasked — it is the original arc-tuned segmentation mask.
  - F140W mask (1121 HST pixels) covers the arc PLUS the deflector core (the photometry tool's aperture geometry pulled the core in); the two masks have zero spatial overlap.
  - Consequence: F140W is valued for its *image* (sharper Sérsic fit), never for its mask. Using the F140W mask as an arc mask would destroy the deflector-core signal. ### §2.1.1b Principled (reproducible) arc masking — NEW 2026-05-29

The hand-painted masks are now backed by an objective, reproducible recipe (so a referee can reproduce the arc selection), validated + applied to all four bands. [`NOTES_arc_mask_verification_2026-05-29.md`, NOTES_photometry_mask_systematics_2026-05-29.md; scripts/arc_mask_verification.py, mask_method_comparison.py, photometry_systematics.py]

- Locator = F200LP (highest contrast on the blue z=1.302 source). Two independent objective selectors reproduce the F200LP hand mask: (i) a color cut m_F200LP−m_F140W (arc bluer than the red deflector); (ii) a Sérsic-residual mask (subtract a 2D deflector model, flag pixels whose positive residual exceeds k·σ_sky, where k is the detection threshold in units of the sky noise σ_sky and σ_sky is the robust background scatter; we adopt k = 3). On F200LP the Sérsic-residual mask reproduces the expert photometry to 0.016 mag, R_e to 3%.

- Transfer to other bands by reprojecting the F200LP footprint (WCS + `map_coordinates`). Reproduces the expert JWST aperture mags to 0.01–0.02 mag.

- WCS registration (NEW 2026-06-10). A raw WCS reprojection leaves a 0.09–0.18″ residual offset between the F200LP arc and the JWST/F140W frames (HST UVIS↔IR and HST↔JWST astrometry; the JWST i2d GWCS differs from the FITS-header WCS astropy reads). On the thin arc that mis-covers one edge and lets a companion slip past the mask. We cross-correlation-register the reprojected F200LP arc (and the F140W color layer) to each band's own image near the deflector (`photometry_systematics.register_shift`), which drives the residual offset to 0.00″ (verified F140W 0.179→0, F150W2 0.154→0, F322W2 0.089→0). Field-galaxy contamination check after registration: no unmasked source (>8σ) falls inside the photometry aperture in any band. *Do not* anchor on the F200LP deflector centroid instead — it is arc-biased (the faint blue deflector centroid is pulled ~0.26″ by the arc), so the arc cross-correlation is the right fiducial.

- Deflector model = single-component Sérsic (per user 2026-06-10). The deflector is an elliptical, so a single Sérsic is the physically-motivated light profile — we do not fit a bulge+disk decomposition (an earlier 2-component model was only a numerical kludge to make the residual-based mask growth behave; see the mask-evolution note below). The single Sérsic is fit per band with masked pixels excluded, capturing 66–87% of the box flux (F140W is the weakest, with a localized central residual from the absent bulge/disk split, but that residual sits inside the arc-free core and does not enter the mask).

  – *Implementation gotcha (documented so it isn't reintroduced):* `astropy.modeling.Sersic2D`'s `amplitude` parameter is I(r_eff), not the central peak. Since I(0)/I(r_eff)=exp(b_n)≈2150 at n=4, initialising `amplitude` at the data peak over-predicts the centre ~2000× and LevMar collapses the amplitude to ≈0 (the whole galaxy then reads as positive residual → catastrophic over-masking). Initialise `amplitude ≈ peak·exp(−b_n)` (~peak·0.05 for n≈3) and bound it to (peak·1e-3, peak·1.5), exactly as `sersic_total_photometry.fit_sersic2d`.

- IR extends the source footprint — grown band-specifically (HST color gate / deep-JWST morphology guard): the deeper/sharper JWST bands see the arc's fuller extent, so we region-grow the F200LP arc seed into the IR. The "is this pixel source, not deflector body?" test is gated two ways depending on the band's depth (a candidate must also be a >k·σ_sky residual, within R<5″, and spatially contiguous with the arc seed — region growing via `scipy.ndimage.label`):

  1. HST bands (F200LP, F140W) — COLOR GATE: the pixel must be bluer than the deflector core by ≥ ΔC = 0.5 mag in F200LP−F140W (`DCOLOR`), at k = 3 (`K_EXT`).
  2. Deep JWST bands (F150W2, F322W2) — MORPHOLOGY GUARD: the pixel must be source-dominated, `data − sky − model > MORPH_FRAC·model` (excess beats the single-Sérsic deflector model → it is OUTSIDE the Sérsic-dominated region), at the lower k = 2 (`K_EXT_JWST`). No HST color is required, so the deep JWST data captures the diffuse arc emission HST is too shallow to color-confirm. Because the single Sérsic already accounts for 66–87% of the deflector flux, `resid > model` rejects the deflector body (model-dominated) while admitting the diffuse source (excess-dominated) — verified: raw mag stable (no deflector loss), 0 unmasked >8σ sources in the aperture, the deflector core <0.4″ never masked. F150W2 IR-extension grows 1049→5288 px (the diffuse arc), F322W2 415→1230 px.

  – Why the color gate is needed (and is the right physics). A *single* Sérsic, being the correct elliptical model, still under-fits the bright extended IR galaxy body slightly, leaving a low-level positive residual over the *red* galaxy. Residual-only growth (criterion 1 alone, the old 2-component recipe) therefore sweeps red galaxy-body light — and even neighbouring companions — into the "arc" mask (verified: F150W2 mask balloons and raw mags fall 1–2 mag). The z=1.302 source is a blue, star-forming lensed galaxy; the deflector is a red, passive elliptical. Requiring the grown pixels to be bluer than the deflector core (criterion 2) exploits this intrinsic 4000-Å-break contrast to keep the blue source extension while rejecting the red body. The mask thus becomes a property of the source color, not of the (imperfect) deflector residual. This is the same physical contrast used by the validated F200LP color selector (i) above; we are simply applying it per-pixel during the IR growth.

  - Color construction. Per target-band grid, reproject the F200LP and F140W surface brightness (`scripts.arc_mask_verification.reproject_intensity`, WCS + bilinear `map_coordinates`) and form −2.5·`log10(SB_F200LP/SB_F140W)` (arc → more negative). Constant ZP offsets cancel because we threshold on the contrast vs the deflector-core color (median in a 0.4–0.8″ annulus), not on an absolute color.
  - Caveats. (a) *(RESOLVED 2026-06-10)* the HST color gate is conservative where HST is shallow and was missing faint diffuse JWST arc emission (user catch via the deep-stretch `_diffuse_mask_check`); the deep-JWST morphology guard above now captures it using JWST's own sensitivity + the source-dominance test, without over-masking the deflector (raw mag stable). (b) $\Delta C = 0.5$ mag and MORPH_FRAC = 1.0 are tunables; the resulting masks reproduce the expert hand-mask $M_\star$ (11.33), our calibration that they neither over- nor under-mask. (c) The morphology guard assumes the single Sérsic captures the deflector majority — true here (66–87%); if a future target has a worse single-Sérsic fit, re-check that `resid>model` still protects the body.

- raw vs fill-in: raw photutils (masked) discards the deflector light under the arc; the single-Sérsic fill-in (replace masked pixels with the model) recovers it. Under the color-gated masks the fill-in correction is −0.19 to −0.27 mag per band (smaller than the old 2-component +0.18 to +0.96 mag, because the cleaner masks remove less deflector light). The headline is the empirical raw value with the best (per-band, color-gated) mask; the fill-in is carried as a one-sided +sys (deflector light the mask removes). PSF effect on the fill stays negligible (`scripts/psf_fill_model.py`: ≤0.004 mag → $\Delta M_\star \ll 0.01$ dex; arc is outside the PSF core).

- per-band vs global mask (union of all per-band masks): masking systematic ±0.086 dex (10%); the raw↔filled (under-arc) term dominates (±0.057), the per-band↔global mask-definition term ±0.028. See §3.2.

- Mask-evolution note (why we settled on the single-Sérsic + color-gated mask). Three attempts, documented in `results/figures/mask_attempts_comparison.png` + notebook 12 §8:

  - (A) 2-component (bulge+disk) Sérsic + residual-only IR growth — over-masks: residual-only growth grabs red galaxy body + a companion (F150W2 15768 px, F322W2 4258 px) → raw biased low (11.16) and a large +0.31 fill-in. The bulge+disk was never physical for this elliptical; it was a kludge to tame the residual growth.
  - (B) single-Sérsic with a broken fit (amplitude→0) + color gate — DISCARDED artifact: the near-zero model made the residual term meaningless and collapsed the fill-in to ~0.01 mag (apparent 11.34 ± 0.013). Caught via the black model panel in the diagnostic figures.
  - (C) single-Sérsic (fixed init) + color-gated IR growth + WCS registration — FINAL: tighter, arc-focused, correctly-registered masks (F150W2 7698 px, F322W2 1997 px; companion now masked), raw central 11.36 consistent with the validated expert hand-mask 11.33, fill real but smaller (−0.19 to −0.28), masking systematic ±0.086 (vs ±0.16 for 2-comp). Physically motivated (single Sérsic) + physically-gated (source color) + astrometrically registered (§2.1.1b).

- Files (in sibling dir `../velocity_dispersion_from_IFU/`): `AGEL020613-011417A_F200LP_WFC3_cutout_L3_mask.fits` (500×500, 25″×25″); the mask was revised 2026-02-28 (`_mask_20260228.fits`); original kept as `_mask_OLD.fits`.

- JWST bands: same arc geometry; the HST F200LP mask is re-projected from the HST WCS onto each JWST frame. [METHODS §II.4]

- For the IFU/σ_e application (context): F200LP mask reprojected to the 0.30″ IFU grid via `scipy.ndimage`.

`map_coordinates(order=0)` nearest-neighbor; 35/154 = 23% of spaxels (28% by I-weight) flagged inside R_e=2.097″ (a geometric LOWER bound — the 1.27″ seeing spreads the arc much wider; see §2.4.1 step 3 + `scripts/psf_mask_fraction.py`). [feedback_masking_strategy.md]

§2.1.2 Estimating R_e

- Headline R_e = 2.097″ = 14.76 kpc at z = 0.67564 (best-mask F140W+F200LP CoG). [CLAUDE.md; METHODS §II.3/A3]
- "Best mask throughout" (2026-06-11): the headline R_e is now the best-mask (single-Sérsic + color/morph gate + WCS registration) curve-of-growth = 2.097″, superseding the expert-mask 2.305″. The best masks genuinely remove outer diffuse-arc + interloper light that slightly inflated the expert-mask R_e; the model-based Sérsic r_eff (1.897″) and sky-subtracted CoG (1.922″) both confirm the inward shift. The HST 2-R_e companion mask is empty (companions live only in deep JWST), so it does not enter the HST R_e — the global color/morph mask alone gives 2.097″ (self-consistent fixed point). `scripts/re_mask_sensitivity.py`, `scripts/Re_bestmask_reconciliation.py`.
- photutils validation (**`scripts/validate_Re_photutils.py`**): our azimuthally-averaged masked CoG is independently reproduced by photutils `RadialProfile` (integrated) to ±0.002″, and unmasked matches photutils `CurveOfGrowth` (direct 2D aperture sum) to ±0.004″. A naive masked aperture-*sum* (photutils `CurveOfGrowth` dropping masked pixels) would bias R_e +0.25–0.41″ high → our azimuthal-fill is the correct masked-CoG treatment. `centroid_2dg` is robust; `centroid_quadratic` fails on extended light (expected — a point-source tool).
- Method systematic (best mask) = ±0.100″ (raw CoG 2.097 / sky-sub CoG 1.922 / Sérsic r_eff 1.897, peak-to-peak/2; `results/Re_cog_reconciliation_bestmask.npz`).
- Method: mean of the F140W and F200LP best-mask curves-of-growth (`scripts/final_sigma_e.py:curve_of_growth`, the PRODUCTION script):
  - best-mask F140W CoG = 1.912″; F200LP CoG = 2.281″; mean = 2.097″ (expert-mask values were 2.168/2.441/2.305).
  - Masked CoG zeros mask-flagged HST pixels before the radial flux integral, so the lensed arc / companions do not bias the half-light point.
  - Centers: HST-mean `photutils.centroid_2dg` centers (same as the σ_e pipeline), both bands.
- Do NOT confuse with **`scripts/measure_Re.py`** → `results/Re_measurements.npz`. That script uses the image-geometric center and produces 10 variants (3 sources × 4 masking strategies: `unmasked`, `zeroed`, `proper`, `PSF-convolved`); its "proper-mask" mean (2.633″) is not the headline — early-reference only. (`measure_Re.hst_Re` previously hardcoded pixscale=0.08 for both bands; fixed 2026-06-08 (A3c/nb14) to read the WCS pixscale + **`r_max_arcsec=6.0`**, now reconciling with the production CoG to <0.04″ — but `final_sigma_e` remains the headline path.)
- Superseded prior values: IFU white-light R_e = 2.61″ (nb05/06 era); expert-mask HST CoG = 2.305″ (pre-2026-06-11). A Ca H+K+G depth-map gives 2.90″ (kinematic-tracer estimator, not the photometric R_e).
- R_e as a σ_e systematic (test D7 → best-mask grid): σ_e rises monotonically with R_e. Re-measured at the wide window over a 7-point R_e grid bracketing the 2.097″ headline (`scripts/run_sigma_e_Re_grid.py`); the adopted R_e-source systematic = ±5.13 km/s (best-mask CoG family {1.912/2.097/2.281″}), folded into the budget (§3.1, §2.4.3). The full grid incl. CaHK+G 2.90″ (±9.98) and CaHK+G's +12.4 km/s deviation are cross-checks, NOT folded. [CLAUDE.md]

§2.1.3 Measuring Fluxes

- Tools: `photutils` aperture photometry (interactive masking GUIs `scripts/photometry_masking_{HST,JWST}.py`). **`synphot`** is NOT currently used — flux→AB is done directly from header zeropoints. *(If the outline wants synphot, that is aspirational; flag with co-authors.)* [METHODS §II.4]
- AB zeropoints — read per-image from FITS headers, never hardcoded. [CLAUDE.md; reference_hst_jwst_data_properties.m
  - HST (drizzled e⁻/s): `ZP_AB = −2.5·log10(PHOTFLAM) − 21.10 − 5·log10(PHOTPLAM) + 18.6921`.
  - JWST (i2d, MJy/sr): `ZP_AB = −6.10 − 2.5·log10(PIXAR_SR)`.
- Four bands / instruments / programs:

Telescope	Instrument	Filter	Program (PI)	Drizzled scale	EXPTIME	ZP_AB	AB (aperture)
HST	WFC3/UVIS	F200LP	16773 (Glazebrook)	0.0500″/pix	600.0 s	27.344	22.613
HST	WFC3/IR	F140W	16773 (Glazebrook)	0.0800″/pix	597.7 s	26.446	19.134
JWST	NIRCam SW	F150W2	05594 (Mahler)	0.03075″/pix	1836.0 s	28.033	18.942
JWST	NIRCam LW	F322W2	05594 (Mahler)	0.06301″/pix	1836.0 s	26.475	18.604

[reference_hst_jwst_data_properties.md] - JWST pointing comes from the cluster program ACT-CL J0206.2−0114 (ACT DR5, Hilton et al. 2021); the deflector sits in that field. No PAM correction (drizzled). Drizzle pixel correlation: empirical noise ~1.3–2× the idealized CCD-equation value — motivates the 10% flux floor (§2.1.4). - 2D Sérsic verification given masking (the morphology cross-check; notebook 08): single-component 2D Sérsic per band (`scripts/measure_Re.py` framework + the 2026-05-11 bound-fix n∈[1.0,6.0], ellip∈[0.0,0.6], 3-init grid — prevented an n=0.30 flat-disk escape on F200LP), extrapolated to total flux:

Filter	AB aperture	AB Sérsic-total	Δmag	n	r_eff (″)
F200LP	22.613	20.672	−1.94	1.42	2.49
F140W	19.134	18.282	−0.85	1.54	1.88
F150W2	18.942	18.154	−0.79	1.40	1.72
F322W2	18.604	17.633	−0.97	1.97	2.03

Aperture under-counts outer-profile flux by 0.8–1.9 mag, but the integrated-mass effect is only +0.07 dex (see §2.1.4 / Results). Caches: `results/sersic_total_photometry.npz`, `results/bagpipes_sersic_refit.npz`. [METHODS §II.6]

§2.1.4 SED Modeling with `bagpipes`

- Tool: `bagpipes` (Carnall et al. 2018, 2019); BPASS+CLOUDY templates (Bagpipes default). Notebook `02_streamlined_Bagpipes_SED.ipynb`. [METHODS §II.5; reference_sed_fitting.md]
- Model: exponential-τ SFH; Calzetti dust (A_V free, 0–2); free metallicity (0–2.5 Z⊙); redshift prior (0.674, 0.676) bracketing the line-fit z. Sampler Nautilus (n_live=400), 500-sample posterior.

```
fit_instructions = {
  "redshift":   (0.674, 0.676),
  "exponential": {"age": (0.1,15.), "tau": (0.3,10.),
                  "massformed": (1.,15.), "metallicity": (0.,2.5)},
  "dust": {"type": "Calzetti", "Av": (0., 2.)},
}
```

- Why a 10% fractional error floor on each band: a conservative, uniform systematic budget that covers both the formal photon noise and the drizzle pixel-correlation underestimate (empirical noise ~1.3–2× the idealized CCD-equation value). [METHODS §II.4/II.7; reference_sed_fitting.md]
- Systematic test to fill in masked pixels = the 2D-Sérsic-total refit: because the aperture excludes arc-masked pixels and truncates the outer profile, we fit a 2D Sérsic per band (§2.1.3), extrapolate to total flux, and re-run Bagpipes with identical priors. This recovers the flux missing from masked/truncated regions and bounds the mass bias. Result: $\log(M_\star/M_\odot) = 11.40\ {+0.11}/{-0.15}$ (Sérsic-total) vs $11.33\ {+0.07}/{-0.09}$ (aperture) → aperture is biased low by ~0.07 dex (+16% in $M_\star$), within the quoted uncertainty. [METHODS §II.6] NOTE (updated 2026-06-10): this analytic-total refit is superseded by the per-aperture single-Sérsic fill-in under the principled color-gated IR-extended masks (§2.1.1b / §3.2): the headline is the empirical raw value 11.36 (10%) with a one-sided +0.14 dex fill-in systematic reaching 11.47. The 11.40 analytic-total sits just inside that bracket and the central is consistent with the validated expert hand-mask 11.33. The

enhanced comparison figure `results/figures/nb08_sersic_vs_aperture.png` now overlays, for both the empirical-aperture and Sérsic-total choices, the best-fit Bagpipes model SED, the measured photometry, and the filter-convolved model photometry.

- Posteriors (older expert-aperture, now a cross-check): log(M★/M⊙)=11.33 +0.07/−0.09; mass-weighted age 5.10 Gyr [3.69–6.11]; τ = 0.68 Gyr; Z = 0.89 Z⊙; A_V = 0.82 mag; z_phot = 0.675 [0.674–0.676]. Per-band residuals: F200LP +4.7%, F140W −11.3%, F150W2 −8.0%, F322W2 +9.6% (mild single-τ-SFH NIR-slope tension; not catastrophic, within ±0.07–0.15 dex). [METHODS §II.5]
- Caches / figure: `results/bagpipes_sed_results.npz` (the file Figure 2 loads), `pipes/posterior/AGEL0206/0206_real.h5` (delete to refit), `results/bagpipes_sersic_refit.npz`.

---

§2.2 Spectroscopic Analysis

§2.2.1 Calibration — @Keerthi / Kaustubh

- Instrument: Keck II / KCWI, KCRM red arm, Red-Low (RL) grating, Medium slicer, 2×2 binning; dithered at PA 0°/45°/90°. [METHODS §I.1; reference_kcwi_data_properties.md]
- Cube (headline): `Nov17_2025_DESJ0206_RL_combined_mtwdo_icubes_wcs.fits` — the NEW **`_mtwdo_`** reduction (Master Twilight + Dome hybrid flats), K. R. Gupta's latest re-reduction. The earlier no-`_mtwdo_` cube is the OLD reduction (second-reduction cross-check; co-aligned to <0.002″, differs only in flat-fielding; σ_e differs ~7 km/s — the "reduction-pass" systematic).
- Reduction chain: raw frames → `kcwidrp`; stacked with `ISMgas/kcwi/kcwiReduction` by K. R. Gupta (Kaustubh); Dec-29 RED frames pre-reduced by Yuguang Chen. *(Calibration prose is Keerthi/Kaustubh's to write — these are the facts on hand.)*
- Cube geometry: shape (3317, 100, 100); 0.300″ spaxels; ~30″×30″ FoV; λ = 5625–8941 Å at $\Delta\lambda$ = 1.0 Å/pix; BUNIT = FLAM16/arcsec^2.
- Three confirmed nights (totals ≈ 180 min RED + 198 min BLUE on source):

Night (UT)	Program (PI)	RED	BLUE
2024 Dec 29	U204	4×300 s	1×1320 s (excluded by reducer)
2025 Aug 30	K409 (PI TBD)	12×300 s	4×990 s
2025 Nov 17	U002 (Jones)	20×300 s	5×1320 s

FLAG: the cube header (DATE-OBS 2025-11-17, PROGPI Jones) describes only the first input frame — cite the multi-night combine, not the header. K409 PI and Aug-30 DIMM seeing still TBD. - Spectral resolution / LSF: FWHM_inst = 2.355 × DISPSCAL(0.2940) × 1.0 Å = 0.692 Å → R ≈ 10000 at the fit band; σ_v,inst ≈ 12.6 km/s observed (7.5 km/s rest-frame) — the ~270 km/s dispersion is resolved by ~20×. LSF robustness ≤ 0.83 km/s over DISPSCAL ×0.5–×2.0. - Seeing: guide-camera FWHM = 1.27″ (`GUIDFWHM`, Nov-17; conservative).

§2.2.2 Redshift Measurement

- Method — single/multi-line Gaussian fits to the integrated-spectrum features on NIST air rest wavelengths, per-line z inverse-variance combined (`scripts/redshift_verify.py`, notebook 04).

Deflector absorption features (systemic z = 0.67564 ± 0.00033; 6 fits):

Feature	Ion/species	rest λ_air (Å)	obs λ_air (Å)	z_line	σ (Å)
Ca K	Ca II	3933.66	6591.79	0.67574 ± 0.00096	11.5
Ca H	Ca II	3968.47	6649.61	0.67561 ± 0.00097	16.6
Hδ	H I	4101.74	6871.80	0.67534 ± 0.00058	3.3
G-band	CH (band)	4304.40	7209.31	0.67487 ± 0.00160	11.3
Hβ	H I	4861.33	8149.30	0.67635 ± 0.00082	2.8

Feature	Ion/species	rest λ_air (Å)	obs λ_air (Å)	z_line	σ (Å)
Ca H+K	Ca II doublet	3933.66 / 3968.47	—	0.67567 ± 0.00069	11.2
Weighted				0.67564 ± 0.00033	

Source emission features (lensed arc, z = 1.30263 ± 0.00003; [O II] doublet, red cube):

Feature	rest λ_air (Å)	obs λ_air (Å)	z_line
[O II] 3726	3726.03	8579.2	1.30357 ± 0.00005
[O II] 3729	3728.82	8585.6	1.30185 ± 0.00005
[O II] doublet			1.30251 ± 0.00004

- Deflector systemic z = 0.67564 (supersedes AGEL DR2 z = 0.67511; ~95 km/s offset, used only in old notebooks). [METHODS §I.2; reference_redshift_verification.md]
- Other absorption features in `ABSORPTION_LINES` (Hγ 4340.47, Fe4383 4383.55, Mg I/Mg b 5175.27, Fe I 5270.40, Na D 5893.00) are NIST-air but were NOT used in the systemic fit (out of the red-cube range at z=0.676, or blended); listed for completeness in `scripts/redshift_verify.py`.

  Air↔vacuum audit (2026-06-11 — PASS). (1) Every rest wavelength is the NIST air value, verified to <0.01 Å (Ca K 3933.66 vs NIST 3933.663, Hβ 4861.33 vs 4861.333, [O II] 3726.03 vs 3726.032, etc.). G-band 4304.40 is the CH molecular bandhead (a blend centroid, not a single atomic line) → it is the most discrepant/uncertain line (z=0.67487, ±0.0016) and is down-weighted by its error. (2) The KCWI cube is vacuum (FITS `CTYPE3='WAVE'`; `AWAV` would denote air) and is converted to air before fitting via the Ciddor (1996)/Morton (2000) IAU-standard index of refraction $n = 1 + 8.34254\times10^{-5} + 2.406147\times10^{-2}/(130-s^2) + 1.5998\times10^{-4}/(38.9-s^2)$, $s = 10^4/\lambda\_vac$ — the same formula the σ_e ppxf pipeline uses (`bootstrap_ppxf.py`; Ciddor 1996). (3) Thus z = λ_obs,air/λ_rest,air − 1 combines both wavelengths in the air frame → internally consistent, z unbiased. Had we left the cube in vacuum and used air rest lines, z would be biased high by ≈ (1+z)·(n−1) ≈ $+5\times10^{-4}$ (≈ +90 km/s); the conversion removes this. - Double verification with ppxf: the Gaussian line-fit z is independent of, and consistent with, the ppxf V_sys (ppxf returns z via $z = (1+z_0)\cdot\exp(V/c) - 1$; Cappellari 2023, the ppxf relativistic velocity convention — *eq. number not yet verified, do not assert "eq. 5c"*). - Source z = 1.30263 ± 0.00003 — measured from the [O II] λλ3726,3729 doublet in the red cube (notebook 04; obs 8579.2/8585.6 Å), consistent with AGEL DR2 z = 1.302. (The ±0.00003 is the formal doublet-fit error; the per-line spread 1.30185–1.30357 is the realistic uncertainty.) - Source rest-UV resonance lines (arc ISM; **`scripts/source_uv_redshift.py`**, 2026-06-11). The bright star-forming arc shows the classic Mg II / Fe II resonance absorption: - Mg II λλ2796.35,2803.53 (red arm): strongly detected (16.7σ / 18.4σ), z(absorption) = 1.30113 ± 0.00003, i.e. a −195 km/s centroid offset vs the systemic [O II] z=1.30263 (both in the *same* red cube, so it is a real *relative* offset, not cross-arm calibration). We do NOT interpret this as an outflow without further work — for a resonance line on a *lensed* arc the offset is degenerate among: (i) Mg II emission infill filling the red absorption wing (P-Cygni; pulls the centroid blueward with no bulk motion), (ii) differential lensing / aperture — the Mg II absorption (continuum-weighted) and [O II] emission (H II-region-weighted) sample different source regions that the lens positions/magnifies differently, and (iii) the blended-doublet single-Gaussian centroid on an asymmetric profile. Report it as an observed Mg II–[O II] velocity offset, with outflow as one (unconfirmed) possibility. - Fe II λλ2344.21,2382.77 (blue arm): detected (7.9σ / 3.0σ; Fe II 2374 only 1.2σ), but at z ≈ 1.306, ~+470 km/s from the red systemic. Because Fe II sits in the separate blue cube (an older `AWAV`/air reduction, ≠ the red `WAVE`/vacuum mtwdo cube), this offset is dominated by a blue↔red wavelength-calibration difference, so the Fe II *velocity* is unreliable cross-arm — the detection is real, the velocity is a `[TODO: cross-calibrate the blue arm]`. (Rest λ are Morton 2003 vacuum; the blue cube is converted air→vac for the fit. NOT a clean independent z — quote Mg II / [O II].) - Companion ellipticals — a galaxy GROUP at the deflector redshift (notebook 04a, `scripts/redshift_verify.bootstrap_redshift`, N=200 wild bootstrap on the red cube). Two early-type companions flank the primary deflector, both at z ≈ 0.676 = the deflector z (physically associated, not interlopers):

Companion	position	sep. from deflector	z (absorption)	z (emission)
NE (“2 o’clock”, larger)	IFU (x,y)=(61,64)	≈ 4.5″ ≈ 31 kpc NE	0.67580 −0.00067/+0.00062	0.67565 −0.00045/+0.00060
SW (“7 o’clock”, compact)	IFU (x,y)=(38,45)	≈ 4.6″ ≈ 32 kpc SW	0.67588 −0.00034/+0.00047	0.67558 −0.00048/+0.00071

Both companions sit within |Δcz| ≲ 60 km/s of the deflector (0.67564) → AGEL J020613−011417 is the central galaxy of a compact group/triple at z ≈ 0.676. (These are the field ellipticals masked in the aperture photometry; their group membership confirms they are not background interlopers.) - Peculiar velocities: we do not correct for peculiar velocities — we use the line-fit systemic z directly (see gap G3; this sentence needs to be added to the draft). - Wavelength convention: AIR rest wavelengths throughout (NIST-verified to <0.01 Å); KCWI native vacuum (`CTYPE3='WAVE'`) converted to air via Ciddor 1996 — see the air↔vacuum audit box above. Do not mix SDSS vacuum values (~1 Å ≈ 80 km/s systematic at 4000 Å). [feedback_wavelength_convention.md]

§2.2.3 Stellar Kinematics & Velocity Dispersion

Spatial regime / masking / R_e

- Headline σ_e is measured on a single I(r)-weighted 1-D aperture spectrum at R < R_e — the co-add-then-fit method of Cappellari et al. 2006 (SAURON IV, §2.3), the literature-standard σ_e (KH13 / Greene+20 / SAURON / ATLAS3D / MaNGA convention). R_e = 2.097″ = 14.76 kpc (best mask; §2.1.2).
- Center: HST-mean `photutils.centroid_2dg` of F140W+F200LP cores, propagated through the KCWI WCS; adopted RA = 31.55613°, Dec = −1.23819° (ICRS). F140W↔F200LP offset 0.36″ (F200LP shifted by the UV-bright arc; F140W is the clean bulge centroid).
- Spatial arc mask: F200LP segmentation mask reprojected to the IFU grid (nearest-neighbor); 38/184 spaxels inside R_e flagged (~27% by I-weight). F140W mask not used (covers the core).
- I-weight map: per-spaxel collapse of the IFU 6500–7500 Å white-light band; spectra summed (not averaged) so the wild bootstrap preserves per-spaxel noise.
- R < R_e/8 (JFK95 σ_c) was dropped — at 0.288″ it sits inside seeing FWHM/2 = 0.64″ (~3 spaxels). No central σ is quoted; σ_e(<R_e/2) ≈ 225 km/s is reported as a gradient diagnostic only.

ppxf setup

- ppxf v9.x (Cappellari & Emsellem 2004; Cappellari 2017, 2023), moments = 2 (first two velocity moments V, σ). [METHODS §I.5/I.6; scripts/bootstrap_ppxf.py]
- Polynomial degrees: additive Legendre `degree` swept over 15–29 (15 values) and pooled; `mdegree = 0`. σ-vs-degree is flat within the bootstrap envelope (no LOSVD absorption by the polynomial). *(Note: METHODS §I.5 prose mis-labels these “multiplicative” — the code passes `degree=…, mdegree=0`, i.e. additive. Describe them as additive.)*
- Three SPS libraries, pooled at the posterior level: FSPS, EMILES, XSL (8–9 templates each over rest 3500–5000 Å).
- Per-SPS native wavelength frame: FSPS = vacuum, EMILES = air, XSL = air. Galaxy frame matched per-SPS (scalar-median vac↔air via Ciddor 1996, following Cappellari’s pattern). Why: the legacy “convert galaxy to air for all three” produced a spurious −90 km/s FSPS V_sys; the fix collapsed the V_sys split-track ~110 → ~15–18 km/s (σ shift ≤2 km/s). [reference_ppxf_vacair_handling.md]
- Per-SPS V_sys subtracted before pooling (σ is V-invariant; a single global V_sys would inflate σ_e by ~10–15 km/s in the discrete cross-check). All three libraries now converge to the same redshift — at the headline R_e=2.097″ aperture the per-SPS V_sys are FSPS −15.7 / EMILES −19.1 / XSL −14.9 km/s → z = 0.67555 / 0.67553 / 0.67556 (a ~4 km/s, Δz≈3×10^{-5} spread; mean z_ppxf = 0.67555 ± 0.00001). This convergence is *the* signature that the air↔vacuum frame fix worked: the legacy “convert all three to air” left FSPS (the vacuum-native library) ~90–110 km/s discrepant from EMILES/XSL; per-SPS native framing collapses it to the ~4 km/s seen here (residual folded into the ±5 km/s frame systematic).
- Emission-line handling (sigma clipping / removing emission lines):
  - Custom no-Balmer goodpixels (`_determine_goodpixels_no_balmer`): masks only the forbidden lines ([O II], [O III], [O I], [N II], [S II]) and keeps Hδ, Hγ, Hβ IN the fit (they are absorption in

this passive deflector; masking them discards ~120 stellar pixels). Hδ: keep unmasked — RESOLVED (M12/nb16): Hδ is well-fit (local-MAD not an outlier), and masking it shifts σ_e by +6–8 km/s = LOSVD *information*, not contamination (the M9 "it's signal" pattern); the `bootstrap_ppxf.py` TODO is closed.
  - z = 1.302 source-emission masks (`ARC_MASKS_REST`, mapped by (1+z_s)/(1+z_l)=1.374): Mg II 2796/2803 (def-rest 3835–3855 Å), [O II] 3727/3729 (5115–5135), [Ne III] 3869 (5260–5340), He I 3819 (5237–5253, added 2026-05-27 as test M8).
  - Tellurics — handled by masking, not correction. One band is masked: the O_2 A-band leading-edge residual at obs 7593–7626 Å = def-rest 4525–4545 Å (`ARC_MASKS_REST`). Confirmed telluric, not z=1.302 source, on 2026-05-18 by the same-observed-λ test (the spike sits at the *same obs* λ in the deflector aperture, the sky box, and a 4–8″ off-deflector spectrum). No full telluric correction is applied; the M10 OH/sky bands (below) catch sky-emission residuals.
  - M10 sky-line audit: bad-pixel catalog `BAD_PIXELS_REST` = 35 entries (26 CR residuals + 9 OH airglow/sky bands), all verified non-source via the arc spectrum.
- Wild bootstrap errors: hybrid Rademacher sign-flip × local-residual scaling (75-pixel rolling-window residual rescale, clipped [0.2,5.0]); N = 500 production per (SPS × degree), N = 50 smoke (agree within 1 km/s). Errors = 16/84 percentiles (asymmetric). joblib parallel, BLAS pinned to 1 thread/worker.

σ_e definition & wavelength window

- σ_e definition — the luminosity-weighted second moment of the LOSVD within R_e. (Defined as the radial integral in Gültekin et al. 2009, eq. (1) [verified 2026-06-12]; the relation we place it on is Kormendy & Ho 2013 eq. 3.) Two mathematically-equivalent representations:
  - Analytic / luminosity-weighted integral — Gültekin et al. 2009 eq. (1) (their 1D form; generalised to the 2D axisymmetric area element dA = 2πr dr):

$$\sigma_e^2 = \frac{\int_0^{R_e} (V^2 + \sigma^2)\, I(r)\, 2\pi r\, dr}{\int_0^{R_e} I(r)\, 2\pi r\, dr}.$$

  - Discrete spaxel/bin sum — the same moment evaluated over the N spaxels (or Voronoi bins) i inside R_e, with F_i = the I-band flux (luminosity weight) of spaxel i; the 2πr factor is captured implicitly because the number of spaxels per annulus scales with area:

$$\sigma_e^2 = \frac{\sum_{i=1}^{N} F_i\, [(V_i - V_{\text{sys}})^2 + \sigma_i^2]}{\sum_{i=1}^{N} F_i}.$$

  This discrete spaxel/bin sum is Cappellari et al. 2013 (ATLAS3D XV, MNRAS 432, 1709) eq. 29 — $\langle v_{\text{rms}}^2 \rangle_e \equiv \langle V^2 + \sigma^2 \rangle_e \equiv \sum_i F_i (V_i^2 + \sigma_i^2) / \sum_i F_i$ — equivalently the discretised form of the Gültekin 2009 eq. 1 integral. (Their eq. 29 also carries an inclination/projection element in the dynamical-model framework; our §7 uses the *observed-plane* projected V_n, σ_n directly and omits that term.) It is NOT "Cappellari 2006 eq. 1" — a prior draft mis-cited it there; Cappellari 2006 eq. 1 is the aperture correction $(\sigma_R/\sigma_e) = (R/R_e)^{-0.066}$, and its σ_e (SAURON IV §2.3) is the co-add-then-fit method below. [eq. numbers verified 2026-06-12: Cappellari 2006 eq. 1, Cappellari 2013 eq. 29, Gültekin 2009 eq. 1]

- Both forms are implemented and cross-checked here (do not re-derive). The headline (§6cum/nb09) is the co-add-then-fit method of Cappellari et al. 2006 (SAURON IV, §2.3) — co-add the I-weighted R<R_e spectra into one effective spectrum and fit a single LOSVD with pPXF; the width of that co-added spectrum *equals* the luminosity-weighted second moment above (this is what KH13/Greene+20/ATLAS^{3D} report). The discrete annular sum (Gültekin 2009 eq. 1) is the §7 / nb07c cross-check (`run_gultekin_mc`, F_j = Σ I over each unmasked annulus). The two agree at <1σ: co-add-fit (§6cum, narrow) 267.32 ± 24 vs discrete annular (§7, arc-filtered to R<R_safe=3R_e/4) 256.17 −13.0/+12.7. Full implementation + subtleties in `reference_gultekin_implementation.md`.

- ★ What the headline code ACTUALLY computes (`final_sigma_e.extract_aperture_spectrum` → pPXF; this is the number we report, 267.31). We do NOT evaluate either σ_e² sum or integral above. We form the I(r)-weighted co-added aperture spectrum

$$S_e(\lambda) \;=\; \sum_{i=1}^{N} w_i\, S_i(\lambda), \qquad w_i = \frac{I_i}{I_{\text{tot}}}, \quad I_{\text{tot}} \equiv \sum_{i=1}^{N} I_i, \qquad I_i \equiv 0 \text{ for arc-masked spaxels (mask\_weight=0),}$$

  where i indexes the N spaxels inside R_e ($r_i < R_e$), $S_i(\lambda)$ is the spaxel spectrum, and I_i its 6500–7500 Å observed-band IFU flux (the luminosity weight; arc-masked spaxels carry $I_i = 0$ so $w_i = 0$). The denominator I_{tot} is the single normalisation constant (total in-aperture flux) so $\sum_i w_i = 1$. The choice of weight map I_i is itself a budgeted systematic — the "I-shape" term (±2.27 km/s; §2.4.3, **`run_isource_shape_sweep.py`**). Holding the spaxel selection fixed, we re-measure σ_e with 10 alternative weight maps — (1) the headline 6500–7500 Å IFU band, (2) full IFU white-light, (3–4) HST F140W / F200LP reprojected (raw), (5–6) the same arc-masked, (7–8) their 1-D curve-of-growth annular means, (9–10) their 2-D Sérsic-model fits — and take peak-to-peak/2 of the spread (266.83–271.37 km/s on the NEW cube + M10). So I_i = IFU white-light is the *headline* choice, not an assumption: σ_e moves ≤±2.27 km/s across this whole family of luminosity weightings.

  - PSF resolution of the HST weight maps (**`scripts/ishape_psf_convolved_test.py`**, 2026-06-12): the HST F140W/F200LP weight maps (and the Sérsic maps) are reprojected at native HST resolution — they are NOT convolved to the KCWI ~1.27″ seeing. Convolving them to the KCWI PSF before weighting *raises* σ_e by +7–8 km/s (F140W 266→274, F200LP 268→277), larger than the ±2.27 term — but this is a contamination effect, not a resolution improvement, so it is not folded (sensitivity only). Smoothing by 1.27″ (i) leaks the *masked* arc light into the unmasked arc-adjacent spaxels (worst for F200LP, where the z=1.302 arc is UV-bright) and (ii) up-weights the high-σ outskirts — exactly the arc/outskirt contamination the I-weighting + arc mask are built to suppress. The headline IFU white-light map is already at KCWI resolution AND arc-masked AND deflector-light- weighted (it down-weights the arc-adjacent outskirts), so it is the correct resolution-matched anchor; the raw HST maps (arc light confined to masked spaxels) are the right I-shape members. pPXF then fits a single Gaussian LOSVD (`moments=2`) to $S_e(\lambda)$,

$$S_e(\lambda) \;\approx\; \Big[\textstyle\sum_k a_k\, T_k(\lambda)\Big] \otimes \mathcal{G}(v;\, V, \sigma) \;+\; \text{(additive Legendre poly, deg 15–29),}$$

  and $\sigma_e \equiv \sigma$, the fitted LOSVD dispersion. Because superposing spaxel spectra at different line-of-sight velocities V_i broadens the co-add, the width σ of that single fit *equals* the luminosity-weighted second moment $\sqrt{\langle (V_i - V_{\text{sys}})^2 + \sigma_i^2 \rangle}$ — so it measures the Cappellari 2013 eq. 29 / Gültekin 2009 eq. 1 quantity without ever forming the explicit sum (and with V_{sys} absorbed automatically into the fitted V; §2.4.1). Code: `extract_aperture_spectrum` (the weighted sum) + `bootstrap_ppxf.setup_ppxf_inputs_from_spectrum + ppxf(..., moments=2)`. The explicit discrete sum $\sum_j F_j (V_j^2 + \sigma_j^2)/\sum_j F_j$ is computed only in the §7 annular cross-check (`run_gultekin_mc`), never in the headline.

- Why the integral path is the headline (not the discrete sum): the discrete spaxel-sum is (i) per-spaxel-S/N-limited and (ii) sensitive to arc contamination in the outer spaxels — an *unfiltered* discrete sum over the resolved PowerBins out to ~3.4″ is dominated by the z=1.302 arc (outer bins carry σ ≳ 300–440 km/s, V swings of ±150–270; the σ_e blows up to ~350). The I-weighted aperture auto-suppresses the arc (bright deflector core dominates the weight) and avoids the S/N floor, so it implements the same Cappellari definition robustly. The discrete §7 sum is therefore arc-filtered to R<R_safe and used only as the architecture-independent cross-check / for the σ(R) profile.

- Standard wavelength range (headline): rest 3800–5400 Å (`wR3800_5400_arcmask`) — Ca H+K through Mg b / Fe5270 / Fe5335; 2161 good pixels (~2.6× the narrow window). The fit is constrained by the highest-S/N features — Ca H, Ca K, G-band (3800–4400) — and is consistent across windows.

- Blue edge 3800 Å = XSL template floor (def-rest 3675 Å) + 5% velocity padding + retaining Ca H+K; red edge set by KCWI coverage. Provenance is data-driven, NOT a specific paper. Closest precedent: LEGA-C (Bezanson et al. 2018; van der Wel et al. 2021) at z~0.7. DO NOT cite "Cappellari 2017 recommends extending blueward" — that was a hallucination, retracted.
- Cross-check window: rest 4000–5400 Å (`wR4000_5400_arcmask`, Hβ + Mg b, no Ca H+K) — orthogonal Lick-tradition feature set.
- Narrow legacy window: obs 6500–7500 Å = rest 3879–4476 Å (Ca H+K + G); σ_e = 267.95 ± 30.10.

A few systematics tests to describe (full budget in Results §3.1): I-shape (10 I-weight-map shapes, ±2.27), F200 mask weight (w∈{0,0.5,1}, ±6.65), fit-window (3 windows, ±3.82), frame (vac/air, ±5.0), centering (5-center ±0.4″ sweep, ±4.0), reduction-pass (NEW vs OLD cube, ±3.45). Plus the M9 "DO NOT MASK" finding: the +5–7% bump at def-rest 5193–5204 Å is the Mg b LOSVD wing (signal, not contamination); masking it drops σ_e by 7 km/s.

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§2.3 Lens Modeling

> GAP G1 — not in this repo. The companion-paper (Ferrami et al.) lens-modeling content (multiple lens models, PSF construction, joint shear+subhalo constraints, potential-correction triple-checks, BPL vs point-mass differentiating models) is not present in this kinematics+ photometry repo. Source it from the lens-modeling repo (`~/Documents/AGEL/202509_DESJ0206_modeling/`, `20251112_DESJ0206_Pyauto_PRONTO/`) or the Ferrami draft. The only on-hand pointer: PROJECT_BRIEF.md notes the lens-model enclosed mass combines with (σ, M★, z) to give a 4D constraint on the central BH + bulge scaling relations.

System identity for the summary sentence: AGEL J020613−011417 (DES J0206−01), AGEL DR2; source z = 1.302 (spiral), deflector z = 0.67564.

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§2.4 End-to-end procedures (ordered walkthroughs)

This section spells out the actual step order of each pipeline — what runs, in what sequence, with which script — so the Methods section and a referee can reconstruct every headline number. Numeric results are the M12 headline (`results/PAPER_VALUES.json`).

§2.4.1 Spectroscopic σ_e pipeline — step by step

Driver for the headline: `scripts/run_wide_sigma_e.py --cube new_clean_hei --n_bootstrap 500` (calls `final_sigma_e.load_setup` → `extract_aperture_spectrum` → `bootstrap_ppxf` machinery).

1. Reduction (upstream). Keck II / KCWI, KCRM red arm, Red-Low grating, Medium slicer, 2×2 binning, dithered PA 0/45/90°. The headline cube is the NEW **`_mtwdo_`** reduction (Master-Twilight+Dome hybrid flats, K. R. Gupta), `Nov17_2025_DESJ0206_RL_combined_mtwdo_icubes_wcs.fits`. The pre-`_mtwdo_` cube is the OLD reduction (second-reduction cross-check). Cube axes `[λ, y, x]`; 0.30″/spaxel; seeing FWHM 1.27″ (`GUIDFWHM`); σ_inst ≈ 12.6 km/s; R ≈ 10000.
2. Geometry + R_e. HST-mean centre = `photutils.centroid_2dg` of the F140W+F200LP cores (offset 0.36″; F200LP pulled by the UV arc, so F140W is the clean bulge centroid), propagated through the KCWI WCS to a sub-pixel IFU centre. R_e = mean of the F140W (1.912″) + F200LP (2.281″) best-mask curves-of-growth = 2.097″ (`final_sigma_e.curve_of_growth`, r_max=6″; photutils-validated ±0.002″).
3. Spatial arc mask → IFU grid. The F200LP `_mask.fits` (2512 HST px, arc-only) is reprojected onto the 0.30″ IFU grid with `scipy.ndimage.map_coordinates(order=0)` (nearest-neighbour) → `arc_spax_mask`. Mask coverage within R_e (PSF caveat; **`scripts/psf_mask_fraction.py`**): this *geometric* mask flags 35/154 = 23% of spaxels (28% by I-weight) inside R_e=2.097″. But the arc light is spread by the 1.27″ KCWI seeing, so the geometric fraction is a lower bound: growing the mask by the PSF (binary dilation) raises the masked fraction to 71% / 81% (by FWHM/2 = 0.64″) or 96% / 98% (by the full FWHM = 1.27″)

— i.e. a hard seeing-grown mask removes nearly the entire aperture and is infeasible. (Convolving the binary mask with the PSF and cutting at >25% / >10% arc-light contamination gives 42% / 75% by count; the >50% cut, 1%, is a thin-arc dilution artifact.) This is precisely why we instead (i) I-weight the aperture — the bright deflector core dominates, the arc-contaminated outskirts are down-weighted — and (ii) carry the mask-weight systematic (±6.65 km/s, w=0→1; §2.4.3) to bracket the residual arc dilution rather than attempt a hard PSF-grown mask.

4. I-weighted aperture extraction. Build the 6500–7500 Å white-light I-map; the R<R_e aperture spectrum = Σ_spaxel cube · w, w = I-weight with arc spaxels hard-masked (mask_weight=0) and re-normalised. Spectra are summed (not averaged) so the bootstrap preserves per-spaxel noise. (`extract_aperture_spectrum`, `final_sigma_e.py`.)
5. Spectral masking (goodpixels). (a) no-Balmer mask — forbidden lines only ([O II], [O III], [O I], [N II], [S II]); Hδ, Hγ, Hβ kept (absorption; M12/nb16 confirmed keep-unmasked).
   (b) z=1.302 source-emission masks `ARC_MASKS_REST` (Mg II, [O II], [Ne III], He I 3819) + the O_2 A-band telluric. (c) `BAD_PIXELS_REST` = 35 entries (26 CR residuals + 9 M10 OH/sky bands). All in deflector rest-frame Å.
   - Outlier rejection / sigma-clipping decision: ppxf's in-fit `clean=False` — `clean=True` rejects 0 pixels here (the KCWI noise array is overestimated vs the actual residual scatter), so it is inert. Instead, cosmic-ray/bad pixels are flagged by a 3σ local-MAD clip (rolling 75-pixel window, |residual|/local-MAD > 3σ) on the canonical residuals → 52 CR-like spikes, curated into the 26 CR entries of `BAD_PIXELS_REST`; the 9 M10 sky bands were flagged separately at >2.5× median sky-noise. Caveats (carried as a stated sensitivity, not a budget term): the 3σ threshold is not strongly motivated vs alternatives, and cleaning shifts σ_e by only +1.4 km/s (below stat 1σ); the same pixels replicate the shift on the OLD cube (M6) → real CR residuals.
6. Per-SPS frame-aware ppxf. For each of FSPS / EMILES / XSL: match the galaxy frame to the SPS native frame (FSPS=vacuum, EMILES/XSL=air; scalar-median vac↔air via Ciddor 1996), run `ppxf` (Cappellari) with moments=2, additive Legendre **`degree`** swept 15–29 (15 values), `mdegree=0`, over the wide window (rest 3800–5400 Å). Subtract the per-SPS V_sys before pooling.
7. Wild-bootstrap errors. Per (SPS × degree), N=500 hybrid wild bootstrap (`bootstrap_ppxf.compute_local_residual_scaling`): residuals r = galaxy − bestfit; a rolling 75-pixel window estimates the local (wavelength-dependent) noise scale s(λ) (clipped to [0.2, 5.0]); each iteration applies a Rademacher ±1 sign-flip to the locally-rescaled residual, adds it back to the best fit, and re-fits → one σ draw. Seeds are deterministic (`BOOT_SEED=42 + 50000 + 10000·W_IDX + 100·sps_idx + degree_i`) so caches bit-reproduce. joblib `loky`, BLAS pinned to 1 thread/worker; N=50 smoke before N=500 production.
8. Pool + budget. Concatenate all σ samples (3 SPS × 15 degrees × 500) → 16/50/84 percentiles = central + asymmetric stat. The pooled width already marginalises SPS + polynomial degree. Add the §2.4.3 systematic components in quadrature → σ_e = 267.31 −12.98/+12.61 (sym ±12.79) km/s (at the best-mask R_e=2.097″ aperture; cache `results/run_wide_sigma_e/resys_best_mean/`).

§2.4.2 Photometry / M★ pipeline — step by step

Drivers: `scripts/photometry_systematics.py` (masking + photometry), `bagpipes_sersic_refit.py` (SED), orchestrated/displayed in notebook 12. Recipe = F200LP locates, IR extends, fill-in recovers.

1. Locate the arc on F200LP (best source contrast): a 2D Sérsic deflector model is subtracted and a pixel is flagged as arc where its positive residual exceeds k·σ_sky — i.e. k is the detection threshold in units of the sky noise σ_sky (so k=3 means residual > 3σ_sky), with a parallel S/N floor (the pixel's own S/N > SNR_thresh). We adopt k = 3 (`arc_mask_verification.py`): this objective Sérsic-residual mask reproduces the expert hand mask to 0.016 mag, R_e to 3%. k = 3 is the clean saturation point of a 2-D sweep over the two thresholds (SNR_thresh ∈ {2..20} × k ∈ {2..8}) — below k≈3 it grabs noise, above it the masked-flux fraction stops changing.
2. Reproject the F200LP footprint to every band (WCS + `map_coordinates`) — reproduces the expert JWST aperture mags to 0.01–0.02 mag.
3. IR-extend with a COLOR GATE. In the deeper/sharper JWST bands, region-grow the arc seed into pixels that are (a) significant positive residual above the single-Sérsic deflector model (data−model > 3σ_sky), (b) bluer than the deflector core by ≥0.5 mag in F200LP−F140W, and (c) contiguous with the seed within

R<5″. The color gate (b) is essential: a single Sérsic slightly under-fits the bright IR body, so residual-only growth would sweep the *red* galaxy body and companions into the mask; the *blue* z=1.302 source vs *red* passive deflector contrast keeps only genuine source pixels. See §2.1.1b for the detailed rationale + caveats.

4. Deflector model = single-component Sérsic — physically motivated for the elliptical (per user 2026-06-10); not a bulge+disk decomposition. Fit gotcha: `Sersic2D.amplitude` is I(r_eff) not the peak — init ~peak·exp(−b_n), bound to (peak·1e-3, peak·1.5), else LevMar collapses to ≈0. PSF-convolved fill quantified at ≤0.004 mag (`psf_fill_model.py`) — negligible.
5. Two photometry estimators per band/mask: raw `photutils` aperture (masked → discards under-arc deflector light → biased low, mask-size-dependent) vs single-Sérsic fill-in (replace masked pixels with the model → recovers it). Fill-in correction −0.19 to −0.27 mag under the color-gated masks (smaller than the old 2-component +0.18–0.96 mag — cleaner masks remove less).
6. AB magnitudes from per-image header zeropoints (never hardcoded): HST `PHOTFLAM/PHOTPLAM`, JWST `PIXAR_SR`. Four bands: F200LP, F140W (HST WFC3), F150W2, F322W2 (JWST NIRCam).
7. Bagpipes SED (`02_streamlined_Bagpipes_SED`): exponential-τ SFH, Calzetti dust (A_V 0–2), free Z (0–2.5 $Z_\odot$), z prior (0.674,0.676); Nautilus sampler n_live=400, 500-sample posterior. A 10% fractional flux floor per band covers photon noise + the drizzle pixel-correlation under-estimate (empirical noise ≈1.3–2× CCD-equation); 20% is run as a conservative variant.
8. M★ budget = 8 fits {per-band, global} × {raw, filled} × {10%, 20%} (script-reproducible via `scripts/mstar_masking_budget.py` → `Mstar_masking_systematic.npz`, `photometry_systematics_Mstar.npz`) → headline (empirical, raw-central with the best per-band color-gated mask, one-sided +sys, user choice): log M★ = 11.36 ± 0.08 (stat) +0.14 (sys) at 10% [11.26 ± 0.14 +0.18 at 20%], fill-in reach 11.47; explicit masking-approach systematic ±0.086 dex (was ±0.16 under the 2-component masks). Central consistent with the validated expert hand-mask 11.33.

§2.4.3 Systematic-audit procedures (how each budget term was measured)

Each σ_e systematic is a dedicated sweep re-using the §2.4.1 machinery, varying ONE axis and quoting peak-to-peak/2 (or half-Δ for two-point axes), all on the headline cube + masks:

- I-shape (±2.27): 10 alternative I-weight maps (`run_isource_shape_sweep.py`) × 3 SPS × 15 deg × N=250; peak-to-peak/2 of the 10 central σ_e (266.83–271.37).
- F200 mask-weight (±6.65): down-weight the arc spaxels at w∈{0,0.5,1} × 3 SPS × N=500; (w00 269.69 − w100 256.39)/2. *(This is the mask-strength axis. The mask-definition axis — expert/sersic/perband/global reprojected to the IFU grid, `run_sigma_e_mask_systematic.py` — gives ±4.58, which overlaps this; larger-of-two is kept, not added.)*
- Fit-window (±3.82): 3 windows (wR3800_5400 / wR4000_5400 / w6500_7500) × 3 SPS × N=500; peak-to-peak/2 (269.66 / 268.58 / 276.22).
- Frame (±5.0): structural — vac/air per-SPS native-frame choice.
- Centering (±4.0): 5 perturbed HST-mean centres (±0.4″ sweep; `NOTES_centering_investigation`).
- Reduction-pass (±3.45): half-Δ between the NEW and OLD cleaned cubes (269.62 vs 262.72); 2 reductions only — refine if a 3rd lands.
- R_e-source (±5.13): σ_e re-measured over a 7-point R_e grid bracketing the 2.097″ best-mask headline (`run_sigma_e_Re_grid.py`). Adopted = best-mask CoG family {1.912/2.097/2.281″} peak-to-peak/2 = (269.99−259.73)/2. The full grid incl. CaHK+G 2.90″ (±9.98) and CaHK+G's +12.4 deviation are cross-checks, NOT folded (CaHK+G is a different I-map definition, not an error on the photometric R_e). Was ±6.13 at the old expert-mask headline.

Quote the budget table (§3.1) as the result; the iterative history that *led* to this masking + component set is the internal audit trail (M8→M12) at the end of this document — not for the manuscript Methods section.

§2.4.4 Reproducibility / infrastructure

- Architecturally-independent estimators (must not share intermediate state): §6cum cumulative I-weighted single-ppxf aperture (the Cappellari 2006 co-add-then-fit method = headline) · §7 discrete Gültekin (2009) eq. 1 annular spaxel-sum (`run_gultekin_mc`, arc-filtered to R<R_safe) · nb07e arc-spectrum subtraction.

The three agree at <1σ (267.32 co-add-fit vs 256.17 discrete, narrow).

- Caching + seed discipline. Every expensive bootstrap writes a `.npz` keyed by its parameters (results/<sweep>/<params>_N{N}.npz); a cached result is never re-run. Seeds are deterministic (§2.4.1 step 7) so re-runs bit-reproduce; N=50 smoke before N=500 production; joblib with BLAS=1/worker to avoid over-subscription.
- Deterministic values registry (single source of truth). `scripts/paper_values.py` recomputes every headline number from the result caches (with per-entry provenance + formula) → emits `results/PAPER_VALUES.json`. The ApJL figures load that JSON (no hard-coded literals); the summary docs are validated by paper_values.py --check <files>, an anchored, precision-aware drift linter (parses only the canonical `σ_e = N ± M` / `log M★ = N` statements; dated snapshots carry a `pv-skip-file` marker). Regenerate + verify: `python scripts/paper_values.py && python scripts/paper_values.py --check *.md`.

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§3 Results

Figure 2 (left panel) — the integrated spectrum + SED

- Left: KCWI/KCRM Red-Low integrated, I(r)-convolved spectrum of the deflector elliptical, with the ppxf best fit overlaid (and inline residuals — Rodrigo's standard nested-gridspec style). Loads from the wide-window fit + bootstrap pool.
- Right: HST+JWST four-band SED + Bagpipes posterior model. Loads `results/bagpipes_sed_results.npz`.
- Figure file: `AGEL0206_sigma_e_SED_final_wide.pdf` (per outline) / `results/AGEL0206_spectra_SED_fit.pdf`.

§3.1 Velocity Dispersion Measurement

- Final redshifts (to quote in Results). The deflector systemic redshift from Gaussian line fits (6 absorption features, §2.2.2) is z = 0.67564 ± 0.00033, in agreement with the independent ppxf stellar-kinematic redshift z = 0.67555 ± 0.00002 (V_sys = −16 ± 4 km/s relative to the line-fit value; headline R_e=2.097″ aperture). All three SPS libraries converge: FSPS 0.67555 / EMILES 0.67553 / XSL 0.67556 (~4 km/s spread) — the signature of the air↔vacuum per-SPS frame fix (§2.4.1), which collapsed the legacy ~110 km/s FSPS-vs-air offset. The 16 km/s line-fit↔ppxf offset is well within the ±59 km/s line-fit uncertainty (and the ppxf formal ±4 km/s carries an additional ~±5 km/s per-SPS frame systematic). We adopt z = 0.67564 (the line-fit value supersedes AGEL DR2 z = 0.67511). The lensed-source redshift is z = 1.30263 ± 0.00003 (red-cube [O II] λλ3726,3729 doublet). [§2.2.2; `scripts/redshift_verify.py`, `run_sigma_e_Re_grid.py`]
- Headline: σ_e(<R_e) = 267.31 ± 12.79 km/s (asym −12.98/+12.61), the luminosity-weighted σ_e within R_e (Gültekin et al. 2009 eq. 1 definition) measured by the Cappellari et al. 2006 (SAURON IV §2.3) co-add-then-fit method — a single pPXF fit to the I(r)-weighted R<R_e aperture spectrum (R_e = 2.097″ best-mask CoG) — plus the wild-bootstrap error pool. Supersedes 269.62 ± 13.27 at R_e=2.305″ — the "best mask throughout" cascade (2026-06-11) adopts the best-mask R_e=2.097″, lowering σ_e by 2.3 km/s along the rising σ(R) profile (well within errors).
- Reproduce: σ_e at the headline aperture is `results/run_wide_sigma_e/resys_best_mean/` (R_e=2.097″), produced by `scripts/run_sigma_e_Re_grid.py --n_bootstrap 500`; the R_e-source systematic (±5.13, best-mask CoG family) is from the same grid (`results/sigma_e_Re_grid_N500.npz`).

R_e measurement and checks (for the §3.1 paragraph): R_e = 2.097″ = 14.76 kpc, the F140W (1.912″) + F200LP (2.281″) best-mask-CoG mean (photutils-validated ±0.002″); supersedes expert-mask 2.305″ and IFU-only 2.61″; masked CoG removes arc/companion bias.

σ_e systematic budget — *generated by `scripts/paper_values.py --render`; do not hand-edit inside the markers.*

Component	± km/s	Note
stat (N=500)	4.51	asym −5.04/+3.98; pooled 3 SPS × 15 deg (marginalizes SPS + degree)
I-shape	2.27	10-shape peak-to-peak/2 on NEW cube + M10
F200 mask	6.65	(w00-w100)/2; larger-of-two vs mask-approach 5.85 (no double-count)
frame (vac/air)	5.00	carried constant
centering	4.00	carried constant
fit-window	3.82	peak-to-peak/2 across 3 fit windows
reduction-pass	3.45	half-Δ between NEW and OLD reductions
R_e-source (best-mask grid)	5.13	peak-to-peak/2 across best-mask CoG family {F140W 1.91“, mean 2.10”, F200LP 2.28“} at the 2.097” headline (user 2026-06-11; CaHK+G & full grid are cross-checks, not folded)
TOTAL (sym)	12.79	quadrature (sys ±11.96 ⊕ stat)
TOTAL (asym)	−12.98 / +12.61	preserves stat-side skew

Cross-checks (not added): arc-mask-definition ±4.58 (overlaps F200 mask) · full-grid R_e-source ±9.98 (incl. CaHK+G, conservative ceiling) · CaHK+G(2.90″) deviation +12.38 km/s.

(Per-component sweep detail — the per-window / per-estimator values behind each ± — is in §2.4.3.)

- No separate SPS line is folded in: the SPS-library spread collapsed from ~26 km/s at the narrow window (FSPS 253.0 < EMILES 267.5 < XSL 279.7) to ~4 km/s at the wide window (FSPS 271.3 / EMILES 267.2 / XSL 270.4; `PAPER_VALUES.json:per_sps_p50`) — a key argument for the wide window; it now sits inside the pooled stat width (between-SPS ±2.04).
- Quadrature is deliberately conservative (components not strictly independent).
- R_e-source systematic IS now folded in (±6.13, M12 2026-06-08): re-measured at the wide window (nb15) and included in the budget above — the narrow-window ±16.9 is superseded. The full 4-estimator spread was adopted (user decision); the CaHK+G 2.90″ estimator drives it (+10 km/s = rising σ(R), see below), while the light-R_e family alone gives ±2.50.

Systematics paragraph order (per outline): first R_e and I(r) (I-shape ±2.27, R_e-source flagged), then the spectrum (mask ±6.65, frame ±5.0, window ±3.82, reduction ±3.45, centering ±4.0).

Cross-checks (narrow window, architecturally independent, superseded for headline but valid):

- §6cum cumulative I-weighted aperture ppxf: 267.32 ± 24; σ_e(<R_e/2) ≈ 225.78; σ_e(<R_e/8) ≈ 209.18.
- §7 discrete Gültekin annular sum (equal-N inside R_safe=3R_e/4): 256.17 −13.0/+12.7.
- §7b flat-σ extrapolation: 274.37 −16.2/+17.4.
- nb07e arc-spectrum subtraction matches §6cum bit-identically (<1 km/s).
- Narrow Ca H+K + G window (nb09d): 267.95 ± 30.10 — headline within ~2 km/s.

Headline evolution (mask-audit arc): 268.98 (clean) → 271.87 (+He I, M8) → 269.62 (+M10 sky audit) → ±11.77 (M11 re-derivation) → ±13.27 (M12, +R_e-source ±6.13, 2026-06-08). He I pushes σ up ~2.9, the 9 sky bands pull it down ~2.2 (near-cancel); central value 269.62 stable across M10→M12 (only the error grew).

Compare with Greene (2016/2020) and KH13 choices: σ_e(<R_e) here uses the luminosity-weighted single-aperture σ_e (Cappellari et al. 2006 SAURON IV §2.3 co-add-then-fit; Gültekin 2009 eq. 1 definition), exactly as adopted by Kormendy & Ho 2013 (M•–σ eq. 3) and Greene et al. 2020 ARA&A (fig. 5). We dropped R<R_e/8 (JFK95 σ_c) because seeing under-resolves it. For a pure elliptical, galaxy R_e ≈ bulge R_e, so KH13's "bulge R_e" equals the measured total R_e — no bulge/disk decomposition needed.

σ_e physical-meaning discussion: the deflector is a dispersion-supported elliptical bulge; σ(r) should *decrease* gently with radius — any rising σ(r) in raw output is a continuum-dilution artifact from the z=1.302 arc, not a physical gradient. Resolved kinematics (nb11): 17 central spaxels at S/N≥5, σ ∈ [144, 251] km/s (median 201), declining with R, no coherent rotation above noise.

§3.2 Stellar Mass Estimate

- HEADLINE: log(M★/M⊙) = 11.47 +0.09/−0.15 (aperture-corrected total, matched 2 R_e aperture, 10% flux floor; `scripts/aperture_matched_photometry.py`, `results/aperture_matched_photometry.npz`). Total-light mass: empirical flux in a matched 2 R_e elliptical aperture (same physical region all 4 bands; single-Sérsic deflector model; color/morph-gated WCS-registered arc mask + companion mask), corrected to total by adding the Sérsic model's beyond-aperture light (the GAMA/Taylor+2011 `fluxscale` approach; additive variant). Methodologically: Taylor et al. 2011 (GAMA aperture→total for M★), Sonnenfeld et al. 2013 (SL2S lens-deflector Sérsic photometry, mask arc + interlopers), Graham & Driver 2005 (Sérsic total formalism).
- Report FIVE estimators (empirical → model), M★ for each, at 2 R_e: | estimator | log M★ | what it adds | |———–|——–|————–| | raw (empirical aperture, masked) | 11.22 | nothing — lower bound | | raw + apcorr (most-empirical total) | 11.35 | model wings only (masked interior NOT filled) | | filled (within aperture) | 11.36 | model fill of masked pixels | | total (aperture-corrected) — headline | 11.47 | fill + wings | | Sérsic-total (pure model) | 11.41 | everything modeled | The total converges across 1/2/2.5 R_e (11.45–11.49) → the aperture correction is internally consistent. raw converges 2↔2.5 R_e (11.22).
- Sérsic-only (full-model) systematic budget = ±0.19 dex (`scripts/sersic_total_systematic.py`; elliptical multi-start fit, 2026-06-11): stat ±0.12 ⊕ sys ±0.15, with arc-mask choice ±0.125 dominant, model-form ±0.057, Sérsic-fit-n ±0.027, flux-floor ±0.023, apcorr↔pure-model ±0.010. (Was ±0.17 with the circular fit; the mask term grew because the elliptical model widens the expert↔global Sérsic-total gap.)
- Apcorr chain validated vs established codes (2026-06-11, **`scripts/validate_apcorr_established.py`**; petrofit/statmorph absent → photutils + scipy incomplete-gamma references): (1) b_n Ciotti&Bertin99 vs exact `gammaincinv(2n,½)` ≤0.05%; (2) the Sérsic total-flux formula (Graham&Driver05 eq.4) vs a numeric render-to-20 R_e ≤0.03%; (3) the aperture correction — rendered-model enclosed light vs the analytic Sérsic incomplete-gamma law `γ(2n, b_n·2{1/n})/Γ(2n)` (Graham&Driver05) — Δ≤0.0007;
  (4) the empirical hard-edge aperture sum (`in_aperture`) vs photutils `EllipticalAperture(method='exact')` ≤0.09%; (5) the finite-cutout truncation of `sum(model_full)` (the apcorr denominator) vs the to-∞ total ≤0.19% (≤0.002 mag). No production bug; M★ unchanged.
- Single-Sérsic ellipticity fix (2026-06-11). The deflector light model is a single Sérsic per band fit on the best mask. The fitter originally had a single low-ellipticity start that railed into a spurious circular χ^2 minimum (b/a=1 for all bands), under-reporting the shape and biasing the `(1−ellip)` total flux. Fixed with a multi-start (flux-weighted-moment seed à la SExtractor/GALFIT
  - a PA grid + circular fallback, lowest-residual). It recovers the true shape; M★ headline (total) is unchanged at 11.47 (empirical-aperture-dominated). Validated by injection-recovery against astropy's peer-reviewed `Sersic2D` (`scripts/validate_sersic_fitter_synthetic.py`): clean recovery for the deflector regime (n≈1.2–1.6, b/a≲0.85), with a realistic b/a uncertainty ~±0.06 (the formal bootstrap is too tight). GALFIT/imfit/petrofit/statmorph are not installed (conda-env policy).
- The M★ posterior has a longer LOW tail (e.g. total 11.47 +0.10/−0.13). This is NOT the total-Sérsic estimator — it is present in all five estimators including the purely empirical `raw`. It is the age–dust–M/L "outshining" degeneracy in the 4-band Bagpipes fit: the low-M★ tail correlates with young age (r=+0.90), high dust (r=−0.58), and high sSFR (r=−0.83) — young+dusty low-M/L solutions reproduce the same broadband points. We keep the exponential-SFH prior and report the asymmetric posterior as-is (the tail is real model uncertainty, not an estimator artifact).
- Per-band single-Sérsic parameters (appendix table; **`scripts/sersic_parameter_table.py`**):

Band	$\lambda_{\rm piv}$ (Å)	r_e (″)	r_e (kpc)	n	b/a	PA (°)	μ_e (mag/″^{2})	$m_{\rm AB}^{\rm tot}$
F200LP	4972	2.002±0.122	14.09±0.86	1.24±0.15	≳0.95†	—†	25.23	20.92±0.07
F140W	13923	1.765±0.015	12.42±0.11	1.44±0.15	0.86±0.06	4±20	22.36	18.42±0.01
F150W2	16720	1.643±0.005	11.57±0.03	1.22±0.15	0.80±0.06	11±20	21.91	18.29±0.00
F322W2	32470	1.757±0.010	12.37±0.07	1.59±0.15	0.85±0.06	6±20	21.79	17.82±0.01

† F200LP: the deflector is faint in this band, so the single-Sérsic ellipticity/PA are unconstrained (consistent with circular); the IR bands set the shape. Errors are bootstrap, with b/a/PA/n floored by the injection-recovery scatter;

the mask-choice systematic (§2.1.1b) dominates and is separate. The deflector is mildly elliptical (b/a≈0.80–0.86, PA≈4–11° E of N) with low Sérsic index (n≈1.2–1.6) — consistent with the b/a=0.75 adopted for the aperture from the independent light fit. - Aperture provenance / why this supersedes 11.36: the old per-band apertures were hand-drawn in the GUI and inconsistent — F200LP a=1.19″ (~0.5 R_e) captured only 17% of the light vs IR ~46% — so F200LP was under-measured → SED artificially red → M★ biased HIGH (11.36). The matched aperture measures the same region in every band → bluer SED → lower M/L → the empirical raw drops to 11.22. - Flux uncertainties: the *propagated statistical* errors are 0.03–1.6% (F322W2 0.03%, F150W2 0.05%, F140W 0.4%, F200LP 1.6%) — 6–362× below the adopted 10% floor. The 10% is a deliberate systematic floor (zeropoint, aperture, drizzle correlated noise ~1.3–2×, SED-model mismatch), not the measurement error — report the true errors and state the floor (~5% is better-motivated; precedent in GAMA/Taylor+2011, whose error budgets are calibration-dominated, not photon-noise). Floor sensitivity (`results/aperture_floor5_check.npz`): 5%↔10% shifts every estimator's central by ≤+0.04 dex (headline total 11.47→11.50) with tighter stat at 5% — well within the ±0.12 systematic, so the headline is robust to the floor; 10% kept for the headline. - Superseded values (cross-checks): mismatched-aperture 11.36 (F200LP-biased high); 2-component-mask 11.16 +0.31 (over-masked → biased low); expert-aperture 11.33. - "Typical elliptical at z~0.7": M★ ≈ 2.3×10^{11} M⊙ (raw headline) to ≈ 3.0×10^{11} (fill-in upper) at z = 0.6756, passive/quiescent SFH (mass-weighted age ~5 Gyr, low sSFR). Place on the z≈0.5–1.0 mass-size relation (van der Wel et al. 2014; Mowla et al. 2019) and the strong-lens-deflector analog sample (Sonnenfeld et al. 2015, SL2S V — best z-match). [notebook 10 verified citations] - X-ray / quiescence: GAP G2 — no X-ray data in repo. Source externally if you want the "truly quiescent" claim; otherwise frame from the passive SED only.

§3.3 Mass Estimates from Lens Modeling

> GAP G1 — see §2.3. The range of lens-model results, the joint-posterior mass + uncertainty, the BPL vs point-mass differentiating models, M(<r_break), the pedagogical-figure intuition, the small table reproduced from Ferrami et al., the comparison to local galaxies, and the "shape of the mass profile matters" argument are all in the lens-modeling repo / Ferrami draft, not here. The only in-repo pointer is the 4D-constraint sentence in PROJECT_BRIEF.md. Pull these numbers from the companion paper before drafting §3.3.

Key citations (verified in notebook 10)

- ppxf: Cappellari & Emsellem (2004); Cappellari (2017, MNRAS 466, 798); Cappellari (2023).
- σ_e definition: Gültekin et al. (2009) eq. 1 (the luminosity-weighted second-moment integral — verified 2026-06-12); co-add-then-fit method Cappellari et al. (2006, MNRAS 366, 1126, SAURON IV §2.3) [NB: that paper's eq. 1 is the *aperture correction*, NOT σ_e — do not cite "Cappellari 2006 eq. 1" for σ_e]; the discrete spaxel/bin sum (§7) is Cappellari et al. (2013, ATLAS3D XV, MNRAS 432, 1709) eq. 29 (= the discretised Gültekin 2009 eq. 1; eq. 29 also has an inclination/projection term that our observed-plane sum omits) [verified 2026-06-12]; Kormendy & Ho (2013, ARA&A 51, 511) eq. 3 (scatter 0.29 dex); Greene, Strader & Ho (2020, ARA&A 58, 257) fig. 5 [⚠ AUDIT FLAG — see below]; Saglia et al. (2016).
- z=0 IFS calibration: Cappellari et al. (2013, ATLAS3D XV & XX); Krajnović et al. (2013, XVII); Zhu et al. (2024, MaNGA DynPop III).
- Mass–size / analogs: van der Wel et al. (2014); Mowla et al. (2019); Ge et al. (2021); Sonnenfeld et al. (2015, SL2S V) [⚠ AUDIT FLAG — see below].
- Vacuum↔air: Ciddor (1996). Wide-window precedent: Bezanson et al. (2018); van der Wel et al. (2021).
- Source-z catalog: AGEL DR2 (Carleton et al., in prep). Cluster: Hilton et al. (2021, ACT DR5).

Citation audit + referee challenges (2026-06-08)

Citation audit (vs the user's Zotero library; "not in library" ≠ fabricated — most are real foundational papers simply not yet added to Zotero, flagged so the .bib can be completed):

- Verified in-library, metadata exact: Cappellari (2017, MNRAS 466, 798) ✓; Kormendy & Ho (2013, ARA&A 51, 511) ✓; Greene, Strader & Ho (2020, ARA&A 58, 257) ✓ *(metadata only — see flag)*; Hilton et al. (2021, ApJS 253, 3 = ACT SZ-cluster catalog) ✓; Cappellari (2023, MNRAS 526, 3273) ✓. Also in-library and relevant: McConnell & Ma (2013) — a massive-BH M•–σ / M•–M_bulge anchor.
- ⚠ FLAG 1 — Greene, Strader & Ho (2020, ARA&A 58, 257): bibliographic metadata is CORRECT, but this review is titled "Intermediate-Mass Black Holes." It does provide M•–σ / M•–M★ relations (legitimately citable as a *comparison* relation, as Fig. 4 uses it), but for a ~10^9 M⊙ SMBH the *primary* anchors should be KH13 / McConnell & Ma 2013 / Saglia+2016, with Greene+20 as one of the overplotted comparison relations. ACTION: verify the "fig. 5" number (the scaling-relation figure in Greene+20 may not be fig. 5) and ensure the text frames it as a comparison, not the primary massive-BH relation. *(Not a fabrication — a framing/figure-number check.)*
- ⚠ FLAG 2 — Sonnenfeld "SL2S V" (2015): the library contains SL2S IV (Sonnenfeld et al. 2013, ApJ 777, 98), not paper V (2015, ApJ 800, 94). ACTION: decide which you mean and either correct to "IV (2013)" or add the 2015 V paper to Zotero. Do not let the 2013 record stand in for a 2015 citation.
- ⚠ FLAG 3 — Bezanson et al. (2018): no Bezanson-first-author 2018 item in the library. If you mean the LEGA-C velocity-dispersion paper (Bezanson et al. 2018, ApJ 858, 60), add it / verify.
- To add to Zotero before the .bib is complete (real papers, just absent; metadata NOT independently verified here — confirm on add): Cappellari & Emsellem (2004), Cappellari et al. (2006, SAURON IV), Gültekin et al. (2009), Saglia et al. (2016), Cappellari et al. (2013, ATLAS3D), Krajnović et al. (2013), Zhu et al. (2024, MaNGA DynPop III), van der Wel et al. (2014), Mowla et al. (2019), Ge et al. (2021), Ciddor (1996), van der Wel et al. (2021, LEGA-C DR3).

Referee challenges to the current doc (anticipate / address in drafting):

1. R_e method family (was "non-converged"; largely ADDRESSED 2026-06-11). Headline R_e = 2.097″ (best-mask raw CoG, r_max=6″) is now photutils-validated to ±0.002″ (`validate_Re_photutils.py`: independent `RadialProfile` integration), and the raw CoG sits at the top of a tight best-mask method family — sky-sub CoG 1.922″, Sérsic r_eff 1.897″ → method systematic ±0.100″ carried explicitly. The raw-CoG-is-top-of-family caveat remains (no sky pedestal subtraction), but it is now bounded and documented, not a flagged unknown. Knock-on: R_e sets the σ_e aperture and (weakly) M★; both effects are inside their systematic budgets.
2. R_e-source budget term (RESOLVED 2026-06-11). Now uses the best-mask CoG light family only {1.912/2.097/2.281″} → ±5.13 — the referee-preferred light-family treatment. CaHK+G 2.90″ (a different I-map definition) and the full grid (±9.98) are reported as cross-checks, NOT folded, so the budget no longer conflates aperture-uncertainty with σ(R)-physics.
3. Rising outer σ(r) attributed to arc dilution. The claim "σ(r) should decrease; any rise is the z=1.302 arc" is the right physical call, but it is *also* what lets us drop the high-σ outer bins; make the arc-contamination evidence explicit (EW / continuum-dilution diagnostic, nb07e) so it does not read as motivated reasoning.
4. He I 3819 source-emission mask rests on a 3-pixel residual cluster — defensible (consistent across reductions, matches z=1.302), but small; a referee may probe it. The M8/M9/M10 audit trail covers this.
5. σ_inst = 12.6 km/s vs σ_e ≈ 270: well-resolved, not a challenge — but state the LSF treatment.

Flagged open items (carry into drafting)

1. ~~R_e-source systematic not folded in~~ RESOLVED 2026-06-08 (M12, nb15): re-measured at the wide window (±6.13) and folded into the budget → headline ±11.77 → ±13.27.
2. Reduction-pass ±3.45 rests on only 2 reductions — refine if a 3rd lands.
3. ~~Hδ may still need targeted masking~~ RESOLVED 2026-06-08 (M12, nb16): decision = keep unmasked (Hδ is well-fit, not a contaminant; masking it would discard LOSVD information). TODO in `bootstrap_ppxf.py` closed. 3b. NEW (A3c, nb14): raw CoG R_e is r_max-non-convergent; headline 2.305″ is the top of a 2.1–2.5″ family (R_e method systematic ±0.08″), not folded into σ_e/M★ — flagged for a decision.
4. Asymmetric-error tables: older 4-corner/M10 tables print −17.92/+17.65 (pre-M11), M11 was −11.98/+11.57 (±11.77); current (M12) is −13.45/+13.10 (±13.27). Quote the M12 value (canonical: `results/PAPER_VALUES.json`).

5. No X-ray / quiescent data in repo (G2). 6. No lens-model content in repo (G1).
6. "We do not account for peculiar velocities" needs to be written in (G3 — true, just add it).
7. Source-z (1.302) error bar + its cache are a `[TBD]` placeholder.
8. K409 (Aug-30) PI and per-night DIMM seeing still TBD for acknowledgements.

Internal appendix — σ_e measurement audit trail (M8→M12) — NOT for the manuscript

> Superseded by "best mask throughout" (2026-06-11). The numbers in THIS appendix (σ_e=269.62, total ±13.27, R_e-source ±6.13, R_e=2.305″) are the M12 (2026-06-08) state. They were the headline until the best-mask cascade adopted R_e=2.097″ (best-mask CoG, photutils-validated), which moved σ_e → 267.31 ± 12.79 and re-derived R_e-source → ±5.13 (best-mask CoG family; `run_sigma_e_Re_grid.py`). The current state is §2.1.2 / §2.4.3 / §3.1 above. This trail is kept verbatim as M12 provenance; do not quote its numbers as current.

This is the iterative history of how the final §2.4.3 masking + systematic-component set was reached. The Methods section should present the final procedure (§2.4) and the budget (§3.1), not this trail. Kept here as internal provenance / referee-rebuttal backup. Each step is also a `TESTS` row with its own cache.

- M8 — He I 3819 (added as a source-emission mask). A 3-pixel positive-residual cluster at def-rest 5244–5248 Å = source-rest 3818–3820 Å matches He I 3819.6 from the z=1.302 source; consistent across both reductions → astrophysical, not a CR. Masked. σ_e (NEW) 268.98 → 271.87.
- M9 — 5193–5204 Å bump (deliberately NOT masked). A +5–7% feature; checked against the arc spectrum (no source counterpart) and the noise spectrum (no sky counterpart) → it is the Mg b LOSVD wing (signal). Smoke-masking it drops σ_e by 7 km/s → DO NOT mask. (Same "it's signal, not contamination" logic later applied to Hδ in M12.)
- M10 — full sky-line audit. Flag every band with cube `noise_sky` > 2.5× median across the fit window, and cross-check each against the arc spectrum to confirm no source counterpart → 9 OH/sky bands added to `BAD_PIXELS_REST` (26 → 35 entries). σ_e (NEW) 271.87 → 269.62.
- M11 — cube-matched systematic re-derivation. Re-ran the I-shape / mask-weight / fit-window sweeps on the NEW cube + M10 masks (replacing carried OLD-cube values). The fit-window term collapsed ±15 → ±3.82 on the cleaned cube; total sys ±17.16 → ±10.81 (total ±11.77).
- M12 (2026-06-08) — R_e-source fold-in + two cross-checks. Folded the wide-window R_e-source systematic (±6.13) into the budget → sys ±12.43, total ±13.27. Added the arc-mask-definition cross-check (±4.58; overlaps the F200-mask term, not double-counted) and resolved the Hδ open item (keep unmasked — well-fit, masking it discards LOSVD information). Central σ_e 269.62 unchanged across M10→M12; only the error grew.

Net effect of the trail: central σ_e settled at 269.62 km/s by M10 and has not moved since; the error bar evolved ±17.78 (pre-M11) → ±11.77 (M11) → ±13.27 (M12). The mask set (He I + 35-entry sky/CR + no-Balmer) and the 7-component budget that the final §2.4 procedure uses are the M12 result.